

Your average All-in-One water cooler

An AIO generally has three components, the CPU block and the radiator with fans attached. The CPU block takes up a lot less space since all the cooling takes place in the radiator. So let's take a look at the individual components within your average AIO water cooler.

CPU Block

We'll start from the absolute bottom.

The copper plate at the bottom is the actual heatsink which comes in direct contact with your CPU. In order to increase the amount of heat that can be drawn from the CPU, the copper plate has a lot of fins. Since this is a completely sealed environment, debris isn't that big a concern and therefore, the copper plate has an array of fins on the topside. These fins are very thin and have barely any gap between individual fins. So the fin density is extremely high and as a result the net surface area for the coolant to interact with the copper plate is high. This results in a much higher heat dissipation. The coolants used in such AIOs are also of low viscosity, so they can easily pass through between the fins.

On top of the cold plate lies a small shroud. If the coolant were to flow straight down onto the copper plate in a haphazard manner then there would be a lot of turbulence within the chamber and as a result, it would affect the cooling. To ensure proper flow in the right direction, the coolant is channeled from one end to another. This may be from one end of the microchannels to the other, or it may be forced down from the top and then drawn out from the sides. The basic idea is to ensure smooth directional flow so that the freshly introduced coolant draws heat from the copper plate and is on its way out to the radiator in the least possible time.



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